

Alejandro López Salazar

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EDUCATION

- **Rheinische Friedrich-Wilhelms-Universität Bonn** October 2025 - September 2027
MSc. in Mathematics Bonn, Germany
 - Specialization area: Algebra, Number Theory and Logic.
- **Universidad de Complutense de Madrid** September 2019 - July 2025
BSc. in Mathematics Madrid, Spain
 - Specialization area: Pure Mathematics.
 - Bachelor Thesis : H-Cobordism Theorem (in Spanish).
- **Universidad de Complutense de Madrid** September 2020 - July 2025
BSc. in Physics Madrid, Spain
 - Specialization area: Fundamental Physics.
 - Bachelor Thesis : Quantum Implementation of NP Problems modeled as Ising Hamiltonians (in Spanish).

AWARDS/FELLOWSHIPS

- **Departmental Collaboration Scholarship** November 2024 - July 2025
Spanish Government
 - Working in the Department of Computer Architecture and Automation in UCM, focusing on Quantum implementation of NP problems modeled as Ising Hamiltonians. Awarded by the Spanish Government.

TEACHING EXPERIENCE

- **Private Tutor** June 2023 - July 2025
UCM Opportunities
 - Tutoring students from high school and university. In the last case, the taught courses were: Linear Algebra, Ring and Group Theory, Differential Calculus and Discrete Math.

TALKS/SEMINARS

- Graduate Seminar on "Geometric Invariant Theory. Affine quotients: definition and elementary properties" (Rheinische Friedrich-Wilhelms-Universität Bonn), planned for Summer Semester 2026.
- Frongasse Seminar on "[Revisiting Serre Duality in Derived Categories](#)" (Rheinische Friedrich-Wilhelms-Universität Bonn), Summer Semester 2026.

ATTENDED WORKSHOPS AND SCHOOLS

- **School on "Exponential Motives"** September 2026
Universität Duisburg-Essen
- **Summer School "Geometry of instantons and quivers"** June 2026
Université de Lille
 - Focus on moduli spaces of specific vector bundles—particularly instantons—as well as quiver varieties, cluster varieties, and the rich mathematical and physical connections that link these structures.
- **Kleine AT Workshop** October 2025
Max Plank Institute for Mathematics
 - One-day conference on the Evenness via Spectral Sequences.
- **JAЕ Summer School of Mathematics** July 2024
ICMAT

SKILLS

- **Programming Languages:** Python, Qiskit, Matlab, Java.
- **Languages:** Spanish (Native speaker), English (IELTS: 7), French (Delf B2).